

PROJECT AND TEAM INFORMATION

Project Title

CleanLink: A Community-Funded Cleanup and Micro-Job Platform

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	CleanLink
Team member 1 (Team Lead)	Royal, Raj – 2510370233 2510370233@geu.ac.in 
Team member 2	Khanka, Aditi – 2510380023 2510380023@geu.ac.in 
Team member 3	Batabyal, Lopamudra – 2510370101 2510370101@geu.ac.in 
Team member 4	Nautiyal, Vedant – 2510011197 2510011197@geu.ac.in 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

In cities and nearby towns everyday life produces a lot of solid waste.. Municipal cleanup teams are usually slow, short on staff and respond only after problems arise instead of preventing them. Dumping sites next to markets, bus stops, drains and parks often stay unchecked for weeks. The reason is that there is no system that links the people who see the problem the workers who could clean it and the money required to do it. City authorities also do not have location-specific information about where waste really piles up so they cannot plan cleanup actions, in the best order.

CleanLink fills this gap by launching a community-led cleanup and micro-job platform. Citizens can point out waste spots by uploading pictures. The community and moderators check these reports. The platform then allows businesses, NGOs and private people to fund the estimated cost of cleaning. When money is secured students or gig workers can sign up for the task. Before pictures prove the work is finished CleanLink releases the payment.

This issue matters for three reasons. First uncontrolled waste poses a public health risk. It attracts disease carriers. Pollutes water. Second it hurts the environment. Ruins public areas. Third it hides a chance. Informal and gig workers could earn money from verified cleanup jobs. No easy platform links them to paid, local work.

By merging citizen reports, community funding and verified micro-jobs CleanLink shifts waste management from a passive municipal duty to an active, data-powered community-driven effort. This gives both people and local governments a quicker way to keep neighborhoods clean.

State of the Art / Current solution (1 pt)

Today waste management in Indian towns and cities relies on manual municipal processes. Sanitation staff follow fixed routes and schedules and residents report problems if all through phone helplines, generic complaint apps or word of mouth to local officials. Some cities use apps such as Swachhata-MoHUA or local municipal grievance portals. These apps let citizens file complaints with photos. Waste management typically ends at the report stage. There is no visibility into funding no connection to available workers and no verification loop to confirm that the waste problem was actually resolved.

Crowdfunding platforms such as Ketto and Milaap exist for causes but they are not built for local task-specific problems such as a single overflowing dumping ground. Similarly gig-work platforms such as Urban Company and local labor apps connect workers to paid jobs. They have no waste-reporting or community-verification component built in.

As a result the current waste management solution is fragmented across three systems: complaint reporting, fundraising and gig work. No platform integrates all three, around a verified outcome. That outcome is a location that gets reported, funded cleaned and confirmed clean.

Project Goals and Milestones (2 pts)

Project Goals

The main goal of CleanLink is to create a prototype of a waste cleanup platform that is driven by the community. It will bring together citizen reporting, community fund raising and gig-worker task assignment into one system. On the side the project will show how core data structures such as linked lists, queues, priority queues, hash tables, arrays and sorting, graphs and file handling in C can be used. It will also show how object-oriented design ideas like inheritance, polymorphism, encapsulation and composition in C++ can be applied. Each idea will be used where it fits a need in the system not just as a separate exercise.

Milestones

Milestone 1. Core data and class layer: Build the C data structures like waste spot records, job queues, worker lookup and file storage. Also build the C++ class hierarchy including User, Citizen, Worker, WasteSpot, Job and Campaign. Make inheritance and encapsulation link the two layers correctly.

Milestone 2. Core workflow: Create the flow from report to verify, fund, assign and complete. Add logic to spot reports and to check before and after photos.

Milestone 3. Funding, reputation and dashboard: Put in funding thresholds, a clear funding record that only releases money when work is verified a worker CleanScore that shows worker reputation and a simple dashboard that shows spots reported spots cleaned and money raised.

Milestone 4. Testing and polish: Check edge cases such as reports, campaigns that do not get enough money and disputed verifications. Improve the output and the user interface. Finish the documents and prepare a demo.

By the end the project must show a working simulation of the cleanup life cycle with real data structures and clear reasons, for using each object-oriented concept at every stage.

Project Approach (3 pts)

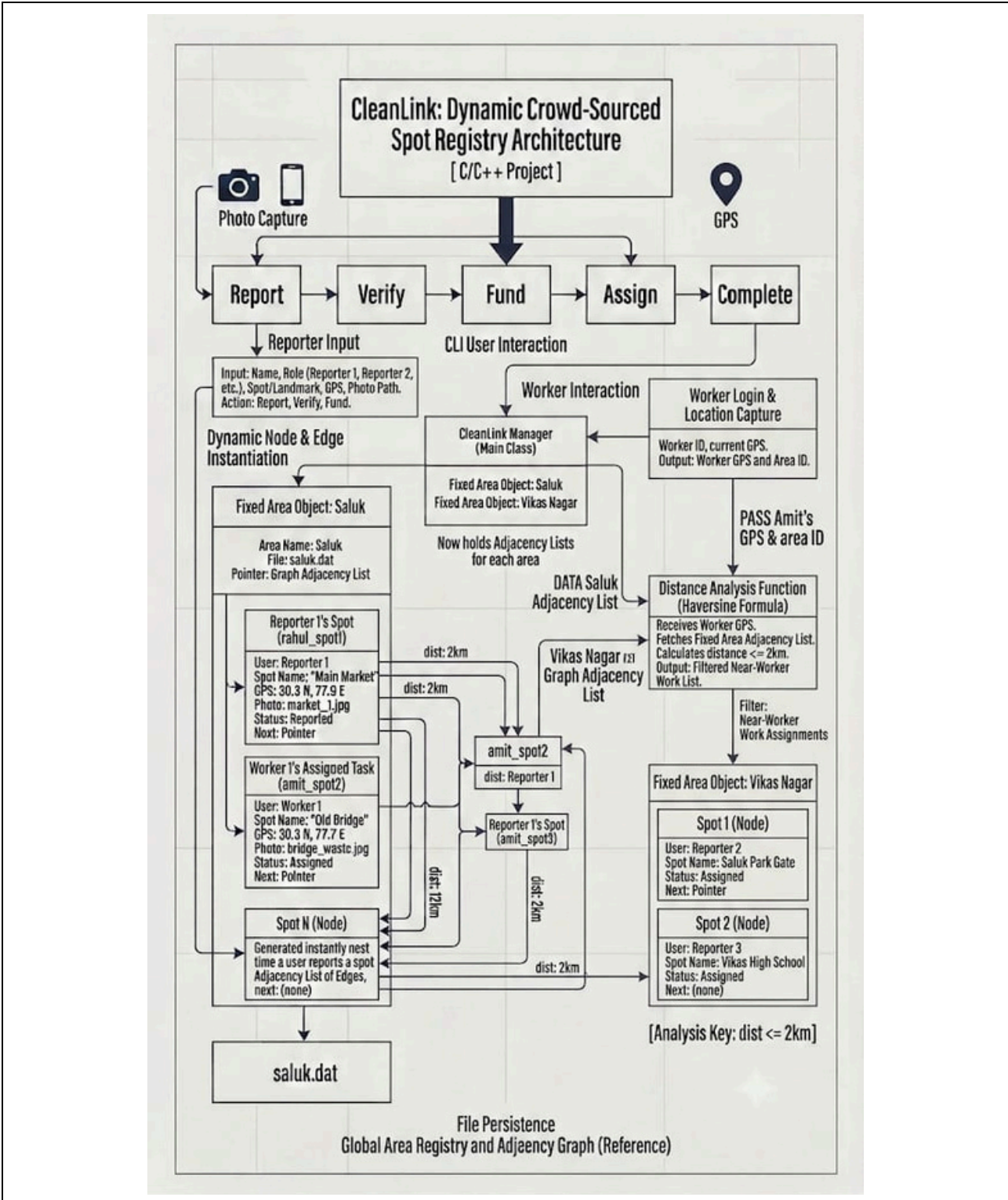
CleanLink will be designed as a two-layer console/desktop application separating the data engine from the application logic. The data engine, written in C, handles all core storage and retrieval operations using fundamental data structures: linked lists for waste spot records, queues and priority queues for job management, hash tables for fast worker and campaign lookups, arrays with sorting algorithms for ranking and searching, a graph structure for proximity-based matching (nearby duplicate reports, nearest available worker), and file handling for persistent storage across sessions.

The application layer provides the cleanlink systems higher control functionality. the primary user class is divided into citizen worker and admin subclasses and each has their own implementation of the common functions. other classes like wastespots and job and campaign provide their respective module specific data services and functionality. some classes as campaign may contain objects of other classes like donations. as a job class may contain payment and assignment class objects. the cpp layer talks to the C data engine via functions. Hence the data storage, algorithms and control logic for the applications are fully decoupled.

the first version of the system will feature a robust command line interface. this was selected because the purpose of this project is to create an elegant user interface and to demonstrate C++ concepts and the data structures. if there is time a simple text menu or a light weight Graphic user interface may be created using ncurses or QT with a stylized console dashboard. this would allow for a more user friendly final demonstration.

This project follows a bottom up approach . first build C data structures and file storage then C++ classes

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The final outcome of CleanLink will be a working prototype that demonstrates the complete waste cleanup process, from reporting a waste spot to verification, funding, worker assignment, and payment release. The project will showcase how C data structures such as linked lists, queues, priority queues, hash tables, graphs, arrays, sorting, and file handling can solve real-world problems, while C++ object-oriented concepts like inheritance, polymorphism, encapsulation, and composition manage the application's logic. The deliverables will include a console-based application, a worker reputation system (CleanScore), funding and campaign records, persistent local data storage, technical documentation, and a final demonstration showing the end-to-end cleanup workflow.

Assumptions

The project assumes that users submit genuine waste reports and that moderators can correctly verify before-and-after cleanup photos. It also assumes that workers complete assigned tasks honestly. Since this is a prototype, payments are simulated instead of using a real payment gateway, and data is stored locally using files rather than a cloud database. These assumptions allow the project to focus on demonstrating the core workflow, data structures, and object-oriented design instead of real-world deployment challenges.

References

- 1.Swachh Bharat Mission (Ministry of Housing and Urban Affairs). <https://swachhbharatmission.ddws.gov.in/>
- 2.Central Pollution Control Board (CPCB). Municipal Solid Waste Management Reports. <https://cpcb.nic.in/>
- 3.Ministry of Housing and Urban Affairs (MoHUA). Swachh Survekshan and Urban Waste Management Resources. <https://mohua.gov.in/>
- 4.Ketto Crowdfunding Platform. <https://www.ketto.org/>
- 5.Milaap Crowdfunding Platform. <https://milaap.org/>
- 6.Urban Company. <https://www.urbancompany.com/>